

Java Script Variables & Data Types – Questions Only (concept Booster)

These questions are arranged from **basic** → **tricky** → **logic-heavy** → **real-world scenarios**.

🌀 BASIC UNDERSTANDING QUESTIONS

1. What is a variable in JavaScript, and how is it different from a constant?
 2. Why was `let` and `const` introduced when `var` already existed?
 3. What will happen if you use a variable without declaring it?
 4. Can a variable declared with `const` ever change? Explain with reasoning.
 5. Is JavaScript statically typed or dynamically typed? Why?
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🌀 `var` / `let` / `const` – OUTPUT & LOGIC QUESTIONS

6. Predict the output:

```
console.log(a);  
var a = 10;
```

7. Predict the output:

```
console.log(b);  
let b = 20;
```

8. Predict the output:

```
var x = 1;  
var x = 2;  
console.log(x);
```

9. Predict the output:

```
let y = 5;  
let y = 10;
```

10. Predict the output:

```
const z = 100;  
z = 200;
```

11. What is **Temporal Dead Zone (TDZ)** and why does it exist?
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🌀 PRIMITIVE DATA TYPES – THINKING QUESTIONS

12. List all primitive data types in JavaScript.
13. Why are primitive data types called **immutable**?

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14. Predict the output:

```
let a = 10;  
let b = a;  
b = 20;  
console.log(a);
```

15. What is the difference between undefined and null?

16. Predict the output:

```
typeof undefined;  
typeof null;
```

17. Why does typeof null return object?

18. Predict the output:

```
console.log(null == undefined);  
console.log(null === undefined);
```

● STRING, NUMBER, BOOLEAN – TRICKY CASES

19. Predict the output:

```
console.log(1 + '2');
```

20. Predict the output:

```
console.log('2' - 1);
```

21. Predict the output:

```
console.log(true + false);
```

22. Predict the output:

```
console.log(10 / '2');
```

23. Predict the output:

```
console.log('10' + 5 - 2);
```

● SYMBOL & BIGINT – ADVANCED THINKING

24. What problem does Symbol solve in JavaScript?

25. Predict the output:

```
Symbol('id') === Symbol('id');
```

26. Why can't Symbol be implicitly converted to string?

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27. What problem does BigInt solve?

28. Predict the output:

```
console.log(10n + 20n);  
console.log(10n + 20);
```

● NON-PRIMITIVE DATA TYPES – LOGIC QUESTIONS

29. What is the key difference between primitive and non-primitive data types?

30. Predict the output:

```
let obj1 = { a: 1 };  
let obj2 = obj1;  
obj2.a = 5;  
console.log(obj1.a);
```

31. Predict the output:

```
let arr1 = [1, 2, 3];  
let arr2 = arr1;  
arr2.push(4);  
console.log(arr1);
```

32. Why are objects and arrays compared by reference and not by value?

33. Predict the output:

```
console.log([] == []);  
console.log({} == {});
```

● REAL-WORLD SCENARIO QUESTIONS

34. Why should configuration values usually be declared using const?

35. A bug appears only in production because of var. How could that happen?

36. Why is using var inside loops dangerous?

37. When should you intentionally use null instead of undefined?

38. Why does JavaScript allow such loose type conversions?

DEEP CONCEPT BOOSTERS (INTERVIEW FAVORITES)

39. Why does `NaN !== NaN`?

40. Predict the output:

```
console.log(typeof typeof typeof 1);
```

41. Why does JavaScript treat numbers as floating-point internally?

42. Why are objects mutable but primitive values are not?

43. What happens in memory when you assign an object to another variable?

CHALLENGE QUESTIONS (DO NOT SKIP)

44. Predict the output:

```
let a = 5;
{
  let a = 10;
}
console.log(a);
```

45. Predict the output:

```
var b = 5;
{
  var b = 10;
}
console.log(b);
```

46. Predict the output:

```
const user = { name: 'A' };
user.name = 'B';
console.log(user.name);
```

47. Predict the output:

```
let x = null;
console.log(typeof x);
```

48. Predict the output:

```
console.log([] + []);
```

49. Predict the output:

```
console.log({} + []);
```

50. Explain **why** JavaScript behaves this way instead of just giving output.

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☒ INSTRUCTIONS

- Solve all questions on paper or in your head
- Do NOT run the code
- When finished, say:

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